Reference, Invariance and Quantized Time — A Dated Record

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Status and provenance

This document records four lines of thought developed by the author in dated working sessions up to 5 November 2026. It is a working record, not a peer-reviewed paper. Author: Joern Christian Anders, B.Sc. Process Technology. Earlier dated records of Parts III and IV (4 and 5 Nov 2026) are listed under References.

- **Author's contributions:** the guiding questions and hypotheses of all four parts; the design, units and naming of the Uniquantsystem (Part IV); the requirement of two independent directions for a fixed reference (Part III).
- **AI assistance:** formalization, literature connections and the numerical runs were carried out with the AI assistant Claude (Anthropic) in those sessions. Established results are credited to the cited literature, not to the author.
- **Timestamp:** the date above is self-declared. An independently verifiable date requires deposit in a public archive that issues a DOI (for example Zenodo) or a preprint server.
- **Status of the claims:** each part separates what is established, what is the author's proposal, and what is still open. The strong form of Part I conflicts with current observations; this is stated in Part I.

Abstract

Four connected results share one rule: define, then measure. **(I)** Cosmic expansion can be moved into a choice of frame, a static space with growing masses. A finite, static universe with constant energy density, however, changes measured invariants — supernova time dilation, T(z), Tolman dimming — and is excluded. **(II)** Under relative observation in strong fields, event probabilities are invariant. Drift arises only when a coordinate-dependent quantity loses its transformation term (Jacobian, Itô or Christoffel term, Radon–Nikodym derivative, lapse). A missing overarching force stays invisible, shows as an anomaly, or is silently absorbed into other parameters; only overdetermination across regimes detects the last case. **(III)** The calibration point 00, the pole of the solar system's invariable plane and the CMB dipole fix all 10 degrees of freedom of a reference frame; quasars check it, and a closure condition attributes any drift to one anchor. **(IV)** The Uniquantsystem counts time in integers on a 1/23 scale; every boundary lies within 0.037 s of a defined anchor, which is proved. An integer lattice of lunar arguments forecasts the Moon over 161

years to 4.1″ against PyEphem, three times better than free circles. The rule transfers to machine learning as reparametrization invariance, error feedback, environment-split validation and structured features.

Part I — Expansion as a description effect

The weak form of the hypothesis holds: cosmic expansion can be moved into a choice of frame. The strong form — a finite, static universe with constant energy density and fixed time — changes measured invariants and is excluded by them.

Hypothesis (author, 5 Nov 2026)

Because space and time are linked, every local description of motion carries a small correction. Summed over very large distances, a continuous description of motion that is not continuous distorts: points appear farther apart, small relative motions appear large, and everything appears to expand. Proposal: define the universe as finite, anchored by known reference points plus a fixed reference tied to the local system; treat particle density as uniform and non-expanding; divide time discretely; describe dynamics only through probability distributions and density.

What holds

1. **Redshift is an integral of local shifts.** Neighbouring comoving observers at distance c dt differ in velocity by H c dt, which gives an infinitesimal Doppler shift (Bunn & Hogg 2009):

$$\frac{d\lambda}{\lambda} = H\,dt = \frac{da}{a} \quad\Rightarrow\quad 1+z = \frac{a(t_0)}{a(t_e)}$$

2. **Cosmic distances are conventions.** Proper, comoving, luminosity (d_L) and angular-diameter distance (d_A) are all legitimate. For the CMB ($z \approx 1100$), $d_L / d_A = (1+z)^2 \approx 10^6$. Recession speeds above c are coordinate artefacts, not motions in the sense of special relativity.
3. **Expansion can be moved into the masses.** Flat FLRW space-time is conformally flat. The rescaling $\Omega = 1/a$ with masses $\tilde{m} = m{\cdot}a$ gives a static Minkowski space in which particle masses grow (Wetterich 2013). All dimensionless observables are unchanged; the dynamics is relocated, not removed.

$$ds^2 = a(\eta)^2\left(-c^2\,d\eta^2 + d\vec{x}^{\,2}\right), \qquad \tilde{g} = a^{-2} g$$

4. **A locally anchored reference** is a set of Fermi normal coordinates along the observer's worldline. It is valid up to the curvature scale, about c/H_0. Beyond it the chart fails, not the physics.
5. **Constant particle density is already standard** in comoving coordinates. What changes measurably is energy density and temperature.

Where the strong form fails

What a choice of coordinates or units can remove is not physics; the invariants remain. A static, finite universe with constant energy density changes these:

Measured invariant	Expanding frame (or equivalent)	Static / tired light
SN Ia light curves stretched by (1+z) (DES 2024)	yes	no time dilation
CMB is a blackbody to better than 10^{-4}	yes	not explained
CMB temperature $T(z) = T_0(1+z)$, seen in distant gas clouds	yes	no
Tolman surface brightness $\propto (1+z)^{-4}$	yes	$(1+z)^{-1}$
Dynamical stability	yes	Einstein static model unstable (Eddington 1930)

Finiteness

- A finite universe without boundary is allowed, for example the 3-sphere S^3, the 3-torus T^3 or the Poincaré dodecahedral space S^3/I^*, where I^* is the binary icosahedral group of order 120.
- Curvature is measured as $\Omega_{tot} = 0.9993 \pm 0.0019$ (Planck + BAO). The small dodecahedral space proposed by Luminet et al. (2003) needs $\Omega_{tot} \approx 1.013$, about 7σ away. A large one remains possible but practically unobservable.
- Searches for matched circle pairs in the CMB found none. Galaxy pair-separation histograms show no lattice peaks (cosmic crystallography). Fixed "corner points" would break the measured isotropy, 10^{-5} after removing the dipole.

Discrete motion

If motion is discrete but described as continuous, errors should accumulate with distance, for example as an energy-dependent light travel time:

```
\Delta t \approx \frac{E}{E_{QG}}\cdot\frac{D}{c}
```

The gamma-ray burst GRB 090510 (Fermi-LAT) gives $E_{QG,1} > E_{Planck}$. Linearly accumulating discreteness is thus excluded up to the Planck scale. Causal set theory avoids this: random, Lorentz-invariant sprinkling accumulates no error, and Sorkin derived $\Lambda \sim 1/\sqrt{N} \sim 10^{-120}$ in Planck units in advance.

Precise form of the author's question

1. **Gauge versus invariants.** Physics is an equivalence class under diffeomorphisms and, possibly, Weyl rescalings. "Expansion" and "growing masses" are two representatives of the same class.
2. **Holonomy.** Frequencies at distant points can only be compared by parallel transport, which is path-dependent in curved space-time. A unique relative velocity of distant objects does not exist. This is the exact sense of "motions that are not motions".
3. **The question becomes:** which quotient space is the physical one? Shape Dynamics (Barbour, Koslowski, Mercati) treats absolute scale as gauge and is closest to the hypothesis. The Wheeler–DeWitt equation with the Page–Wootters mechanism derives time from correlations inside a probability distribution, close to "only distributions and density".

Status

- **Established:** redshift as integrated local shifts; distance conventions; frame equivalence of expansion and growing masses.
- **Author's proposal, open:** a finite reference description with discrete time, formulated on the quotient space.
- **Excluded:** the static variant with constant energy density (SN time dilation, T(z), Tolman test, stability).

Part II — Drift errors under relative observation

The probability of an event does not depend on the observer. Drift arises when a coordinate-dependent quantity is used without its transformation term; the error is then systematic and accumulates.

Questions (author, 5 Nov 2026)

1. If the observation point is relative and strong overarching fields act, does a drift error enter the probabilities?
2. How does this look on Earth, where cosmic forces act and no absolute zero reference exists?
3. If an overarching cosmic force is missing from the equations, does that make another equation wrong?

Invariant and non-invariant quantities

A change of frame is a measurable map f, and the measure is pushed forward: $P'(f(A)) = P(A)$.

Invariant	Not invariant
P(A) for events (counting)	densities p(x), modes, non-linear expectations
quadratic variation (diffusion part)	drift coefficient of an SDE
conserved total flux through a hypersurface	marginals "at time t" (depend on simultaneity)
—	rates per unit time (proper time), particle number (Unruh)

Five mechanisms of drift

1. **Densities — the Jacobian.** Dropping it shifts mode and mean; in curved space-time $\sqrt{|g|}$ enters the normalization.

```
p_Y(y) = p_X\!\left(f^{-1}(y)\right)\left|\det \frac{\partial f^{-1}}
{\partial y}\right|
```

2. **Rates — proper time.** For a static observer, $d\tau \approx (1 + \Phi/c^2)\, dt$, and Poisson rates transform as $\lambda' = \lambda\, dt/d\tau$. Thermal equilibrium is position-dependent (Tolman–Ehrenfest: $T\cdot\sqrt{(-g_{00})} = \text{const}$). GPS clocks drift by about 38 µs per day without correction.
3. **Dynamics — the Itô term.** A change of variables Y = f(X) adds a noise-induced drift $\tfrac{1}{2}\sigma^2 f''$; on a manifold it is the Christoffel term of the Laplace–Beltrami operator. With position-dependent diffusion D(x) and no drift, the stationary density depends on the evaluation point α of the stochastic integral: $p_{st} \propto D^{(\alpha-1)}$.

```
dY = \left(f'b + \tfrac12\sigma^2 f''\right)dt + f'\sigma\,dW
```

4. **Change of measure — Girsanov.** Two observers with equivalent measures see the same paths with different drift θ; the diffusion is unchanged because quadratic variation is defined path by path.

$$dW^Q = dW^P - \theta\,dt, \qquad \frac{dQ}{dP} = \exp\!\left(\int\theta\,dW - \tfrac12\int|\theta|^2\,dt\right)$$

5. **Relativistic and quantum.** "Probability at time t" depends on the chosen space-like hypersurface; only the flux of a conserved current is covariant. For an accelerated observer the particle concept itself depends on the observer (Unruh). This is genuine observer dependence, not an error.

Why it matters. An uncorrected drift b grows as b·t, noise only as σ√t; beyond $t^* \sim \sigma^2/b^2$ the bias dominates. With more data the bias stays while the spread falls as σ/√N: a consistently wrong estimator.

On Earth: no absolute reference is needed

The local inertial frame replaces the absolute reference: an accelerometer reads zero, a gyroscope does not precess. An external potential expands around the observer as follows:

$$\Phi_{\text{ext}}(x) = \underbrace{\Phi_0}_{\text{gauge}} + \underbrace{\nabla\Phi\cdot x}_{\text{removed by free fall}} + \underbrace{\tfrac12\,x^\top T\,x}_{\text{tidal, physical}} + \mathcal O(x^3)$$

- **Order 0** cancels in every normalization: e^(−(E+c)/kT)/Z does not depend on c. The galactic potential ($\Phi/c^2 \approx 6\cdot10^{-7}$) exceeds Earth's own ($7\cdot10^{-10}$) and is still unobservable locally.
- **Order 1** vanishes exactly in free fall (equivalence principle): the orbital accelerations about the Sun ($5.9\cdot10^{-3}$ m/s²) and the Galactic Centre ($2\cdot10^{-10}$ m/s²) drop out.
- **Order 2** cannot be transformed away. Lunar tides reach $1.1\cdot10^{-6}$ m/s², solar tides $5\cdot10^{-7}$ m/s²; galactic tides (~$5\cdot10^{-24}$ m/s²) and Λ in bound systems (~10^{-29} m/s²) are negligible.
- **Rotation is not relative.** Dynamical non-rotation (gyroscopes) and kinematic non-rotation (quasars, ICRF) differ by the geodesic precession of the Earth–Moon system, about 1.9″ per century — a real drift caused only by the choice of cosmic reference.
- **Aliasing.** Sampling once per day at the same clock time folds the lunar tide M2 (1.932 cycles/day) to a period of 1/(2 − 1.932) ≈ 14.8 days, which looks like a slow trend.
- **Selection amplifies frame effects.** The Sun moves at about 370 km/s relative to the CMB ($\beta \approx 1.2\cdot10^{-3}$). The dipole in source counts is D = [2 + x(1+α)]·β (Ellis & Baldwin 1984); the flux threshold adds x(1+α). Measured quasar and radio dipoles exceed the CMB expectation; whether this is systematics or physics is open.

A missing overarching force

True model y = Xβ + g + ε with a missing term g; only Xβ is fitted:

$$\hat\beta = \beta + (X^\top X)^{-1}X^\top g, \qquad r = (I - P_X)\,g + \varepsilon, \qquad g = g_\varnothing + g_\parallel + g_\perp$$

Component	Effect	Consequence
$g_\varnothing$: acts on no observation	invisible	harmless (equivalence principle for a universal, homogeneous force)
$g_\parallel$: in the model's column space	absorbed into other parameters	fit is perfect, parameters are wrong; fails under extrapolation
$g_\perp$: orthogonal to the model	appears in residuals	visible as an anomaly

- Within the calibrated range a missing term is never both invisible and harmful to predictions. Damage arises in the interpretation of parameters and in extrapolation.
- **The only working test is overdetermination:** the same constant estimated through independent channels. The Hubble tension is a current candidate.
- **The same residual has different causes:** Uranus (a missing body, Neptune), Mercury's perihelion (an incomplete law, general relativity), the Pioneer anomaly (systematics, anisotropic thermal recoil).
- **Upward regress is bounded, not excluded.** Unknown physics at a scale Λ appears below it only as renormalized parameters or as corrections of order $(E/\Lambda)^n$. Material-dependent extra forces are bounded at about 10^{-15} by MICROSCOPE.

Status

- **Established:** all mechanisms above are standard results.
- **Author's contribution:** the question in its three steps, which leads to a single diagnostic rule: decompose any shift by Taylor order (offset, mean, covariance, rest) and check, for each order, whether the pipeline absorbs, corrects or amplifies it.

Part III — A three-anchor reference frame

The calibration point 00 plus two physical anchors — the pole of the solar system's invariable plane and the CMB dipole — fix all 10 degrees of freedom of a space-time reference frame; distant quasars check it independently. A frame is drift-free only by

definition; because this one is overdetermined, every drift becomes visible and attributable to one anchor.

Idea (author, 5 Nov 2026)

One direction leaves a circle open: a rotation about it stays undetermined. At least two independent states or directions are needed, and the heavy bodies of the solar system should serve as the base on which the others stand. The precise form of "the heavy bodies as base" is the invariable plane: Jupiter and Saturn carry 61.5 % and 24.9 % of the planets' orbital angular momentum, the four giants together 99.8 %.

What a frame must fix

A complete frame has 10 degrees of freedom, the dimension of the Poincaré group: 4 translations in time and space, 3 rotations, 3 boosts.

Anchor	Fixes	Degrees of freedom	Realization
A: calibration point 00	time origin and origin	4	00 = 12 July 2029, 22:00 UTC; TT; geocentre (GCRS)
D: invariable plane	pole e_3, i.e. 2 of 3 rotations	2	total orbital angular momentum; pole RA 273.853°, Dec +66.991°
C: CMB dipole	rest frame (3) and rotation about e_3 (1)	4	369.82 ± 0.11 km/s towards l = 264.02°, b = 48.25° (Planck 2018)
B: distant quasars	checks all 3 rotations, fixes nothing	0	ICRF3: 4 588 sources, 303 defining; axes to about 10 μas

The triad

The solar system alone cannot supply a fixed direction within the plane, because all perihelia and nodes move. The CMB dipole supplies it:

$$e_3 = \hat D, \qquad e_1 = \frac{\hat C - (\hat C\cdot e_3)\,e_3}{\left\lVert \hat C - (\hat C\cdot e_3)\,e_3 \right\rVert}, \qquad e_2 = e_3 \times e_1$$

Axis	Meaning	RA	Dec
e_3	pole of the invariable plane	273.853°	+66.991°

Axis	Meaning	RA	Dec
e_1	CMB dipole projected into the plane	172.767°	+4.668°
e_2	$e_3 \times e_1$	260.830°	−22.477°

- D and C are 102.57° apart; the dipole lies 12.57° south of the plane. A direction error is amplified only by 1/sin(102.57°) = 1.02.
- **e_3 by definition:** pole of the orbital angular momentum without spins, as in Souami & Souchay (2012). Provisional value from DE405; to be fixed at 00 from DE440 including Ceres, Pallas and Vesta. The Sun's spin is excluded: it is poorly known and would shift the pole by about 2′ (own estimate).
- **e_1 by definition:** Planck 2018 dipole direction, fixed at 00.
- **Accuracy:** e_3 is determined to 0.01–0.3″ depending on the ephemeris and is stable to 0.3 mas over 100 years; e_1 is known to about 0.01°, ICRF3 to about 10 µas.

Residual drift of the physical anchors

Anchor	Residual drift	Cause	Over 966 years
A	ΔT; precession ≈ 50″ per year	Earth rotation, precession of the axis	already carried as series
D	⩽ 0.1 mas (inclination), ⩽ 0.3 mas (node) per 100 years in the ephemerides	angular momentum conservation; only external torques by passing stars	mas range, limited by the ephemeris
B	apparent glide 5.8 µas per year, plus about 30 µas source-structure noise	our acceleration towards the Galactic Centre	≈ 5.6 mas; modelled in ICRF3 at epoch 2015.0
C	velocity changes by about 7 mm/s per year	the same acceleration	≈ 7 m/s; direction at most ≈ 4″

B and C drift from the same cause. That gives the first consistency check: the acceleration Gaia measures from quasars, $2.32 \cdot 10^{-10}$ m/s², must match the change of the CMB dipole.

Why three anchors, and the closure condition

Anchors	What can be learned about drift
1	nothing, there is no comparison
2	that something drifts, not which anchor
3	which anchor drifts, as long as only one does (2-of-3 majority)

The transitions between anchors are defined at 00 so that they close:

```
T_{AB} \circ T_{BC} \circ T_{CA} = \mathrm{id}
```

This is the consistency condition of an atlas on triple overlaps. If the measured loop does not close, the deviation is the drift, and its source shows which transition changed. Every transition is measurable today: A ↔ B by VLBI, B ↔ C as the same acceleration by two methods, C ↔ A through the orbital dipole of the Earth that Planck used for calibration. The two transitions to D are the check numbers of the triad.

Without distortion

The transitions are rigid by definition: rotation, translation and boost, no scaling and no shear. Distortions are not removed but separated, by expanding the field of quasar proper motions in vector spherical harmonics.

Component	Meaning
toroidal dipole	rotation of the frame; the direct test for a cosmic rotation
spheroidal dipole	glide, i.e. acceleration
higher orders	genuine distortion: gravitational waves, lensing, systematics

Use in the 105-tick system

- **Sidereal system longitude:** the angle in the invariable plane, counted from e_1 and fixed at 00, is drift-free. The tropical system longitude $(\lambda - \lambda_0)$ moves with the equinox, about 50″ per year. Their difference, the ayanamsa, is the measured transition A ↔ D.

- **Redshifts** are referred to the CMB rest frame through anchor C; the correction relative to heliocentric values is up to ±0.0012.
- **Checks per epoch:** the closure number, the angle D–C (102.57°) and the rotation of the triad against ICRF3.

Part IV — The Uniquantsystem

A time count in integers only that stays attached to the Sun without accumulating error: every boundary lies within q/2 = 0.037 s of a defined anchor, and any run of units deviates by less than 1 q. The full construction, proofs and studies are in the preprint The Uniquantsystem: Integer Time Counting on a 1/23 Scale (4 Nov 2026); this part summarizes it and adds the results of 5 Nov 2026.

Design (author)

The author conceived and named the system and made its design decisions: the 23-year cycle on the 1/23 scale, the zero crossings with −1 before every zero, quantum irrelevance, the calibration point 00 named jj, the units Jules, joe and Iris, the Noraumlauf and the defined anchor.

Unit	Composition	Length
quantum q	1/23 tick	75/1012 s ≈ 0.0741 s
tick	23 q	75/44 s
cycle	66 ticks + zero crossing of 1 q (2 q with an extra quantum)	1 519 or 1 520 q
day	768 cycles	1 166 595 or 1 166 596 q
year	13 × 28 days + zero day = 280 320 cycles	425 806 080 + J q, J = 1 290 … 1 331
chart / Noraumlauf / great chart	23 years / 7 charts = 161 years / 23^2 = 529 years	residual exactly 0 at each great-chart end
joe	1/1 165 824 lunation = 1.11166″ of mean elongation	29 or 30 q
Iris	1″/299, pole fields on the crust	3.3445 mas ≈ 10.3 cm
Jules	69.159 s, unit of ΔT	5 832 409/6 250 q

Calibration point 00 (jj): Friday 13 July 2029, 00:00 CEST = 12 July 2029, 22:00 UTC. Before jj the parameters are calibrated against observation; afterwards they are definitions, and nature enters only as measured series "nature minus definition".

Anchor and proofs

With Y = 425 806 080 q and x in years since 00, year boundaries are A_n = round(τ(n)) and cycle boundaries B(c) = round(τ(c/280 320)):

```
\tau(x) = Y x + \frac{b\,x + g\,x^2}{529}, \qquad b = 703\,993,\; g = -11
```

1. **Period 529:** the residual ρ_n = τ(n) − A_n repeats exactly after 529 years and is 0 at every great-chart boundary; no shorter period exists because 2g = −22 is coprime to 23.
2. **Residue classes:** at the 23 chart starts the residuals take each value k/23, k = −11 … 11, exactly once, since b ≡ 9 (mod 23).
3. **Quantum irrelevance:** every boundary is within q/2 of the anchor; window errors telescope to less than 1 q; the mean vanishes; residuals are computable and need no storage.

Recomputed here in exact rational arithmetic over 967 year starts: largest residual 0.499 q, RMS 0.284 q (≈ 1/√12, a uniform distribution), mean +0.0009 q, all 23 residue classes, J = 1 331 q in the first year and 1 290 q in 2995.

Method	Largest total error over 966 years	Growth
each year rounded independently	214 q = 15.8 s	linear
stochastic rounding (median of 200 runs)	13.5 q = 1.0 s	∝ √N
cumulative rounding with carried residual	⩽ 0.5 q = 0.037 s	bounded

Computed incrementally this is a first-order error-feedback (sigma-delta) quantizer behind exact integrators; the extra quanta form a Sturmian sequence.

Nature minus definition and further axes

- **Sun:** the true Sun deviates from the anchor by −17.7 to +16.9 min; eight terms of known origin (nutation, Venus, Jupiter, Moon, Mars) reduce the scatter from 6.59 to 0.71 min.
- **Moon (joe axis):** true new moons deviate by up to ±14.1 h from the defined mean lunation; eight terms reduce the scatter from 7.5 h to 3.2 min.
- **Rotation axis:** ascendant, MC and house cusps within 3.1″ of the Swiss Ephemeris over 2029–2264.

- **Polar axis:** the pole is measured, not counted; its forecast is a probability per Iris field (68 % radius 5.5 Ir after 30 days, 14.8 Ir after one year).
- **Tides:** S2 lasts exactly 582 912 q and the spring–neap cycle exactly 582 912 joe, so M2 is the doubled beat between the q and joe grids.

Plato's question: how many circles, and how far do they forecast?

Body	Circles for 1″ in the window	Best forecast over the next 161 years	Best forecast, 322-year window → next 161 years
Sun	24	ellipse + circles 2.5″	ellipse + circles 6.7″ (quadratic elements; 8.3″ linear)
Mars	over 150	ellipse + circles 20.9″	quadratic elements + circles 12.8″
Moon, free circles	107	76.2″ (8.1″ without the 273-year Venus term)	13.6″
Moon, integer lattice D, M, M′, F + Venus term	—	—	4.10″

- **Circles describe, ellipses explain:** in the window the residual keeps falling with every circle; outside it stalls. The right curve replaces more than a hundred circles (Ptolemy → Kepler).
- **A term longer than the window cannot be identified from it:** the 273-year Venus term of the lunar longitude (frequency 18 L_V − 16 L_E − M′) is absorbed as curvature and continued wrongly.
- **Structure generalizes:** the integer lattice uses about a fifth fewer bits than free circles at equal residual and forecasts about three times better. Free circles that know the Venus term still reach only 11.8″, so most of the gain comes from tying all frequencies to four fitted arguments.

Results added on 5 Nov 2026

- **Back-tests (six splits, 322 years fitted, 161 measured, amplitude drift ε fitted from the data):** lunar lattice forecast 4.84″ at K = 60, 4.31″ at 80, **4.11″ at K = 112** (best), 4.18″ at 150. The forecast error does not rise past the optimum (no overfitting), but beyond about 80 circles only the window improves. The 1-SE rule selects K = 101; a 10 % tolerance allows K = 68 (4.50″, 88 numbers fewer).

- **Error floor:** two back-tests stay at about 6″ from K = 50 on, whatever the number of circles (measured spans 1868–2029 and 2029–2190). A floor independent of K points to a missing term, not to missing capacity; the cause is open.
- **Amplitude drift:** the lattice gets an option for linearly varying amplitudes of the terms with M. On a synthetic signal it recovers ε = −0.25 per century exactly; on the oldest split (window around 1224) it finds ε = −0.00234 per century against −0.00236 in Meeus — the shrinking eccentricity of the Earth's orbit, identified from the data. Fitted to two Noraumläufe and measured on the third, the drift lowers the lattice forecast from 4.10″ to 2.34″, and with Venus, Mars and Jupiter to 1.82″. ε fixed to Meeus gives the same (1.81″), so the data determine ε fully. Two free drift numbers per term overfit: better in the window, worse in the forecast (2.71″).
- **Lunar phase:** the phase is D = λ_Moon − λ_Sun. The lattice forecasts the Moon to about 1.8″ (lattice with drift and planets, fitted to the first two Noraumläufe from jj, measured on the third), but the best solar model only to 6.7″, so the Sun dominates the phase error: about 6.9″ ≈ 13.7 s at new moon if the errors were independent. They are not: both share the solar anomaly and the eccentricity factor E, the Sun carries a 29.53-day term from the Earth–Moon barycentre, and common-mode effects such as precession and nutation cancel in D. Aberration does not cancel (about −20.5″ for the Sun), so "apparent" versus "geometric" D shifts the new moon by about 40 s and must be fixed as a convention.
- **Gravitation in the system:** the clock rate L_G (≈ 22 ms per year) belongs to the definition through TT; TDB − TT (1.7 ms) is quantum-irrelevant; a clock 1 km higher gains 79 µs per chart, irrelevant up to about 500 km; a clock on a GPS orbit gains about 4.3 q per chart and needs its own rate. One q per chart corresponds to a potential difference of $9.2 \cdot 10^6$ m²/s².
- **Reference limit:** all values are measured against PyEphem, whose abridged lunar theory is good to about 10″ (≈ 20 s at new moon). The studies therefore show how well the models continue that theory, not the true sky. Calibration against JPL DE441 at 00 is required.

Status

- **Proved:** Theorems 1–3, exact in rational arithmetic.
- **Measured against PyEphem:** all forecast figures; each split is one sample, and K was chosen on the test span, so best values are optimistic.
- **Open:** final calibration at 00 with DE441 and IERS data; the 6″ floor; the forecast as a probability per joe field calibrated on back-tests.

Common thread and transfer to machine learning

All four parts apply one rule: define, then measure. What a choice of description can remove is gauge; what remains is invariant and must be measured; residuals are carried, not discarded; and only an overdetermined system can attribute a drift to its cause.

Principle	Where it appears	Machine-learning counterpart
Gauge versus invariant	I: expansion versus growing masses; II: Jacobian, Girsanov; III: choice of frame	MAP is not reparametrization-invariant, the natural gradient is; a constant logit shift leaves the softmax unchanged
Carry the residual	IV: cumulative rounding, bounded error	error feedback in gradient compression (Karimireddy et al. 2019), compensated summation in low-precision optimizers; stochastic rounding grows as $\sqrt{N}$
Overdetermination attributes drift	II: a parameter estimated across regimes; III: three anchors and the closure condition	domain-split validation, Chow test of coefficients across environments, invariant risk minimization
A missing term hides in other parameters	II: the absorbed component $g_{\parallel}$	shortcuts and confounders: perfect in-distribution fit, failure under shift
Components longer than the window are not identifiable	IV: the 273-year Venus term	long-period structure missing from the training window is absorbed into the trend and extrapolated wrongly
Structure beats free capacity	IV: integer lattice, fewer bits, three times better forecast	structured, sparse dictionaries instead of free Fourier features; minimum description length (Rissanen 1978)
Drift in coefficients	IV: shrinking eccentricity of the Earth's orbit	non-stationary data needs weights with known time dependence, not more fixed weights

Principle	Where it appears	Machine-learning counterpart
Aliasing	II: M2 sampled daily → 14.8-day pseudo-trend; IV: annual sampling	evaluation at fixed intervals under cyclic learning rates or data order creates spurious trends; anti-aliased downsampling (Zhang 2019)
Selection amplifies shift	II: Ellis–Baldwin dipole	a confidence threshold under shift amplifies the bias by the log-slope of the distribution's tail

Diagnostic rule for a training pipeline: decompose every distribution shift by Taylor order — offset, mean, covariance, rest — and check for each order whether the pipeline absorbs it (normalization), corrects it (reweighting by dQ/dP) or amplifies it (thresholding).

Testable predictions and open questions

Parts III and IV make quantitative predictions that can fail; Part I needs one before it can be tested.

Part	Prediction or test	Fails if
I	Any static-frame reformulation reproduces SN time dilation (1+z), $T(z) = T_0(1+z)$ and Tolman dimming $(1+z)^{-4}$	a proposed finite, static description predicts other values for these three
II	Shifts decomposed by Taylor order: offset and mean are absorbed by normalization, covariance needs second-moment alignment	a residual bias remains after mean and covariance alignment without a higher-order cause
III	Closure $T_{AB} \circ T_{BC} \circ T_{CA} = \mathrm{id}$ within measurement error at every epoch	the loop does not close and no single anchor explains it
III	The angle D–C stays at 102.57°, within about 4″ over 966 years	the angle changes by more than the known drift of C
III	Gaia's acceleration ($2.32 \cdot 10^{-10}$ m/s²) equals the change of the CMB dipole (about 7 mm/s per year)	the two methods disagree beyond their errors

Part	Prediction or test	Fails if
III	The toroidal dipole of quasar proper motions is zero	a rotation of the quasar frame against local gyroscopes appears
IV	Every boundary within q/2, every window within 1 q, for all time	cannot fail: proved (Theorems 1–3)
IV	A lattice with amplitude drift for the terms with M lowers the lunar forecast error well below 4″ (against PyEphem already 4.10″ → 1.82″; synthetic 3.3″ → 0.02″)	the remainder against DE441 stays near 4″
IV	Cumulative rounding of weights or gradients keeps the accumulated error bounded, unlike stochastic rounding	accuracy per bit is not better than with stochastic rounding

Open questions

1. **Part I:** what does a finite reference description with discrete time predict differently from ΛCDM, in numbers — for redshift, supernova brightness or the CMB? Without such a number the idea can be neither tested nor credited.
2. **Part III:** the final values of e_3 (DE440 with Ceres, Pallas, Vesta) and e_1 at 00.
3. **Part IV:** calibration against DE441 and IERS data at 00; the cause of the 6″ floor in two back-tests; the convention for D (apparent or geometric); a solar lattice so that the lunar phase is no longer limited by the Sun.
4. **Transfer experiment:** quantize the weights or gradients of a small PyTorch model by cumulative rounding with carried residual and compare accuracy against bits with stochastic rounding.

References

The linked sources were opened in the dated sessions; for the others, check volume and page before deposit.

Records of this work

- J. C. Anders, The Uniquantsystem: Integer Time Counting on a 1/23 Scale, preprint draft, 4 Nov 2026
- J. C. Anders, Drei-Anker-Bezugssystem, working document, 5 Nov 2026

Part I

- E. F. Bunn, D. W. Hogg (2009), The kinematic origin of the cosmological redshift, Am. J. Phys. 77, 688.
- C. Wetterich (2013), Universe without expansion, Physics of the Dark Universe 2, 184.
- A. S. Eddington (1930), On the instability of Einstein's spherical world, MNRAS 90, 668.
- R. M. T. White et al. (2024), DES supernovae: cosmological time dilation, MNRAS.
- Planck Collaboration (2018), Planck 2018 results I (arXiv:1807.06205)
- J.-P. Luminet et al. (2003), Dodecahedral space topology as an explanation for weak wide-angle temperature correlations in the CMB, Nature 425, 593.
- N. J. Cornish et al. (2004), Constraining the topology of the universe, PRL 92, 201302.
- R. Lehoucq, M. Lachièze-Rey, J.-P. Luminet (1996), Cosmic crystallography, A&A 313, 339.
- A. A. Abdo et al. (2009), A limit on the variation of the speed of light from GRB 090510, Nature 462, 331.
- R. D. Sorkin (1991), Spacetime and causal sets.
- J. Barbour, T. Koslowski, F. Mercati (2014), Identification of a gravitational arrow of time, PRL 113, 181101.
- D. N. Page, W. K. Wootters (1983), Evolution without evolution, PRD 27, 2885.

Part II

- R. C. Tolman, P. Ehrenfest (1930), Temperature equilibrium in a static gravitational field, Phys. Rev. 36, 1791.
- J. Dunkel, P. Hänggi (2009), Relativistic Brownian motion, Phys. Rep. 471, 1.
- G. F. R. Ellis, J. E. Baldwin (1984), On the expected anisotropy of radio source counts, MNRAS 206, 377.
- P. Touboul et al. (2022), MICROSCOPE mission: final results of the test of the equivalence principle, PRL 129, 121102.

Part III

- G. Souami, J. Souchay (2012), The solar system's invariable plane, A&A 543, A133
- P. Charlot et al. (2020), ICRF3 (arXiv:2010.13625)
- Gaia Collaboration, S. A. Klioner et al. (2021), Gaia EDR3: acceleration of the Solar System from Gaia astrometry

Part IV and transfer

- J. Meeus (1998), Astronomical Algorithms, 2nd ed., Willmann-Bell, ch. 31, 47 and 49.
- H. Bohr (1925), Zur Theorie der fastperiodischen Funktionen I, Acta Math. 45, 29.
- N. R. Hanson (1960), The mathematical power of epicyclical astronomy, Isis 51, 150.

- J. Laskar (2003), Frequency map analysis and quasiperiodic decompositions, arXiv:math/0305364.
- J. Rissanen (1978), Modeling by shortest data description, Automatica 14, 465.
- S. P. Karimireddy et al. (2019), Error feedback fixes SignSGD and other gradient compression schemes, ICML.
- R. Zhang (2019), Making convolutional networks shift-invariant again, ICML.